

Stage 9 — Is there a remedy that beats masking everywhere?

Where the Shortcut Sits: progress report 9

In one sentence. No remedy dominates ROI masking — none of the seven candidates registered in two rounds (four, then three chosen by a search over twelve ideas) is never worse and better wherever masking fails — but the best ones come within one cell: a class-conditional mean constraint on the masked features (`mask_cmc`) and IRMv1 are better than masking in every Trap A (including a cohort held out from all choices) and on the shortcut-conflicting cases of every unaltered test set, equal to it elsewhere, and lose only on the external ISIC 2020 set (-0.021 $[-0.027, -0.016]$ and -0.009 $[-0.013, -0.005]$ overall AUROC), where every alternative to masking, including the unmasked model, loses.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma, malignant nodule, tumour with a solid component, erosion.
$A \in \{0, 1\}$	artifact present: a real artifact: hair, sonographer caliper, capsule debris.
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ of reversed-test AUROC (Stage 3).
CI	95% crossed seed×image bootstrap interval (Stage 1), 10,000 replicates; one-sided tests at 0.025.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
Candidate	a remedy compared with masking: every contrast is candidate – masking, AUROC.
Dominates	meets every component of the registered criterion (Table 2).
Easy / conflicting / same-artifact pairs	positive–negative pairs whose artifact status follows / conflicts with the association, or is the same in both images (Stage 5).
Development / confirmation	data used to choose a candidate’s settings / data generated only after its registration.

Contents

1 Where this stage starts

3

2	Questions	3
3	Methods	3
4	Experiments	4
5	Results	4
5.1	Round 8: close, but no dominance	4
5.2	The search	6
5.3	Round 9: the same result on new data	6
6	What failed or changed	8
7	Anticipated questions	8
8	Conclusion	9
9	Questions for the supervisor	9

1 Where this stage starts

Stage 6 developed and compared remedies for the failure of masking and concluded that the right remedy depends on what carries the shortcut, and that no remedy improved on masking uniformly on unaltered data. Stage 8 validated the labels the remedies need and showed the failure on a near-benchmark model. One question remained: is there *one* method a practitioner could use instead of masking — never worse, and better where masking fails? This stage answers it with a registered criterion, two registered rounds on data generated only after each registration, and a documented search between them.

2 Questions

Q9a Does any existing or adapted remedy dominate masking by a registered criterion?

Q9b If not, where exactly does the best one lose, and why?

Q9c Can a search over further ideas, including methods from outside medical imaging, find one that does?

Design decisions *The comparison is with masking only.* “Dominates” means not worse where masking already works and better where it fails (Table 2); it does not require beating every other remedy. *Cells where no honest method can win are left out:* easy pairs (a method that stops using the shortcut must lose there) and correlated tests as strong as training (the criterion uses the worse of the reversed and correlated tests). *One criterion, many components.* A candidate dominates only if all 42 components are met (an intersection–union test, so no correction within a candidate); across candidates a fixed sequence, registered in advance, keeps the full 0.025 for the first. *Choices on validation data only.* Every setting is chosen per training set by the worst-group validation AUROC with a guard on the same-artifact pairs, which measure disease ranking whatever the association; masking is always a fallback. *Untouched confirmation.* New trap resamples with seeds used for nothing else, and one cohort (ovary) whose data never influence a choice. The unaltered test sets and ISIC 2020 had been seen before and are semi-confirmatory. *Novelty is not a condition:* every candidate is labelled published or adaptation with its closest prior work; none is new. *At deployment* a candidate may use only the image, its ROI mask and automatic detectors.

3 Methods

Design (no test scoring) Every existing arm was recomputed on one crossed bootstrap (11,135 cells); the closest existing arm met 29 of 36 cells of the criterion. An exact decomposition of the all-pairs AUROC into easy, conflicting and same-artifact pairs showed that balancing loses on unaltered data mainly through the same-artifact pairs, i.e. through disease signal lost by reweighting, not through removing the shortcut.

Round 8 candidates (i) `mask_cmc`: masked features projected onto the complement of the within-class mean differences between images with and without the artifact, so the head may not score them differently on average within a class; a first-moment, linear case of counterfactual invariance (Veitch et al. 2021). (ii) `full_cmc`: the same on the full image. (iii) `mask_bal`: partial group reweighting with a validated strength. (iv) `locrand_loc`: real artifact instances pasted inside and outside the ROI so that presence at each location is independent of the label. Nine further arms (the uniform paste plan, a selector, combinations, AFR, GroupDRO, a soft U-MtE, a label-sensitivity run) are descriptive. Separate claims: controlled sweeps (D5), a fine-tuned paste model (D6), replication with MedSigLIP and ConvNeXt (R). One amendment, before any run, replaced Holm

across candidates by the fixed sequence, made the location-matched paste plan primary and labelled failed components as *loss* or *inconclusive*.

Search After round 8, twelve ideas were tried on development data only (development seeds; validation folds of the unaltered training sets; never a confirmation seed, test set, ISIC 2020 or ovary; enforced in code): backdoor adjustment, product of experts and logit adjustment from NLP and long-tail learning, moment matching, conditional adversarial debiasing, Correct-n-Contrast and counterfactual contrastive learning, V-REx and IRMv1, and combinations. Every attempt is logged.

Round 9 The three best eligible search candidates, in the fixed sequence conditional adversarial debiasing $\rightarrow$ IRMv1 $\rightarrow$ V-REx, with round 8’s `mask_cmc` and `mask_bal` refitted with their frozen recipes as references; new confirmation seeds; registered as the last remedy round.

4 Experiments

Table 1: Experiments of this stage. Each registration was committed before the data it tests were generated (round 8: confirmation seeds 8101–8505; round 9: 9101–9505). Commit times are set by the committing machine.

Experiment	Status	First commit (local time)	Support criterion
Design and scoreboard of every existing arm	post hoc (design; no test scoring of new models)	86e99d8, 2026-09-28 21:19	–
Round 8: four primary candidates (fixed sequence), sweeps (D5), fine-tuned (D6), replication (R)	registered before results (round 8); one amendment before any run	dd5c7ee, 2026-09-28 22:01	dominance: every component of D1–D4 met (intersection–union)
Search over twelve candidates on development data	post hoc (exploratory; development data only)	8645734, 2026-09-29 00:15	ranking only
Round 9: three candidates from the search (fixed sequence)	registered before results (round 9); the last remedy round	6563a66, 2026-09-29 08:44	as round 8

Table 2: The registered criterion for “dominates masking”. A candidate dominates only if every component is met; easy pairs, where any method that stops using the shortcut must lose, are not in it.

Cells	Test (candidate – masking, one-sided 0.025)
D1 all pairs of the six unaltered test sets (thyroid split, capsule, BCN, HAM, MSK, ISIC 2019 $\rightarrow$ 2020)	not worse: lower bound > -0.01
D2 shortcut-conflicting pairs (thyroid, BCN, MSK, ISIC 2020) and Trap A min(reversed, correlated) in four cohorts	better: lower bound > 0
D3 Trap B and clean tests, per cell (16) and pooled over cohorts (4)	not worse: -0.03 per cell, -0.01 pooled
D4 every trap	no shortcut flipping: correlated $\geq$ reversed -0.02

5 Results

5.1 Round 8: close, but no dominance

Table 3. The sequence stopped at its first candidate: `mask_cmc` met 40 of 42 components. It is better than masking in every Trap A — thyroid $+0.330$ [$+0.311, +0.351$], capsule $+0.153$ [$+0.131, +0.177$], dermoscopy hair $+0.193$ [$+0.166, +0.221$] and the held-out ovary $+0.118$ [$+0.089, +0.147$] — and on the conflicting pairs of every unaltered test set, including ISIC 2020 ($+0.078$ [$+0.069, +0.088$]);

it is not worse on Trap B and the clean tests, never flips the shortcut, and is not worse overall on four of the six unaltered test sets (thyroid inconclusive: -0.002 $[-0.011, +0.007]$). Its one loss is ISIC 2020 overall (-0.021 $[-0.027, -0.016]$). `mask_bal` also met every D2 component but lost small amounts overall on thyroid, MSK and ISIC 2020 (Table 4, Figure 1).

Table 3: Round 8 on the confirmation data: no candidate dominates masking. The sequence stopped at its first candidate; the others are reported unadjusted.

Candidate	Components met	All of D2 met	Losses	Fixed sequence
class-conditional mean constraint (<code>mask_cmc</code>)	40 of 42	yes	ISIC 2019 $\rightarrow$ 2020 (D1)	tested, not rejected
mean constraint on the full image (<code>full_cmc</code>)	36 of 42	no	dermoscopy, HAM held out (D1), dermoscopy, MSK held out (D1), ISIC 2019 $\rightarrow$ 2020 (D1)	not tested in the sequence
partial reweighting (<code>mask_bal</code>)	39 of 42	yes	thyroid split (D1), dermoscopy, MSK held out (D1), ISIC 2019 $\rightarrow$ 2020 (D1)	not tested in the sequence
location-matched (<code>locrand_loc</code>)	pasting 39 of 42	no	dermoscopy, BCN held out (D2)	not tested in the sequence

Table 4: Round 8, the two candidates that won everywhere masking fails: the D1 and D2 components and the pooled D3 components (AUROC, crossed 95% CIs); every per-cell D3 component and D4 were met by both.

Component	Cell	<code>mask_cmc</code> — masking	<code>mask_bal</code> — masking
D1 all pairs	thyroid split	-0.002 $[-0.011, +0.007]$	inconclusive -0.008 $[-0.017, +0.001]$
D1 all pairs	capsule	-0.000 $[-0.001, +0.001]$	met -0.001 $[-0.004, +0.000]$
D1 all pairs	dermoscopy, BCN held out	-0.005 $[-0.006, -0.003]$	met $+0.003$ $[+0.000, +0.007]$
D1 all pairs	dermoscopy, HAM held out	-0.003 $[-0.008, +0.002]$	met $+0.003$ $[-0.001, +0.008]$
D1 all pairs	dermoscopy, MSK held out	-0.002 $[-0.006, +0.002]$	met -0.006 $[-0.011, -0.000]$
D1 all pairs	ISIC 2019 $\rightarrow$ 2020	-0.021 $[-0.027, -0.016]$	loss -0.011 $[-0.016, -0.005]$
D2 conflicting pairs	thyroid split	$+0.084$ $[+0.069, +0.101]$	met $+0.048$ $[+0.032, +0.066]$
D2 conflicting pairs	dermoscopy, BCN held out	$+0.036$ $[+0.033, +0.040]$	met $+0.033$ $[+0.026, +0.040]$
D2 conflicting pairs	dermoscopy, MSK held out	$+0.070$ $[+0.059, +0.082]$	met $+0.044$ $[+0.030, +0.058]$
D2 conflicting pairs	ISIC 2019 $\rightarrow$ 2020	$+0.078$ $[+0.069, +0.088]$	met $+0.035$ $[+0.024, +0.046]$
D2 Trap A	thyroid	$+0.330$ $[+0.311, +0.351]$	met $+0.380$ $[+0.355, +0.404]$
D2 Trap A	capsule	$+0.153$ $[+0.131, +0.177]$	met $+0.166$ $[+0.148, +0.185]$
D2 Trap A	ovary (held out)	$+0.118$ $[+0.089, +0.147]$	met $+0.161$ $[+0.135, +0.187]$
D2 Trap A	dermoscopy hair	$+0.193$ $[+0.166, +0.221]$	met $+0.213$ $[+0.187, +0.239]$
D3 pooled	Trap B reversed	$+0.020$ $[+0.014, +0.026]$	met $+0.045$ $[+0.036, +0.054]$
D3 pooled	clean (Trap A models)	$+0.057$ $[+0.048, +0.065]$	met $+0.063$ $[+0.054, +0.072]$
D3 pooled	clean (Trap B models)	$+0.007$ $[+0.002, +0.012]$	met $+0.010$ $[+0.002, +0.017]$

Why ISIC 2020 On ISIC 2020 in-lesion hair still points towards melanoma for most patients. `mask_cmc` removes that shortcut: it gains $+0.078$ $[+0.069, +0.088]$ on the conflicting pairs and loses -0.116 $[-0.139, -0.094]$ on the easy pairs, where the shortcut gives the right answer; the rest of the overall loss comes from the same-artifact pairs.

Separate claims In the controlled sweeps `mask_cmc` and `mask_bal` dominate masking (12 of 12 components each; D5, Holm). The replication with MedSigLIP and ConvNeXt supports the Trap A gain in 5 of 5 cohorts for `mask_cmc` (smallest gain 0.132) and 5 of 5 for `mask_bal`. The paste candidates are not a fair test: in the traps too few images could receive a paste to balance presence between the classes, and a probe separates pasted from real artifacts (AUROC 0.904–0.990); the fine-tuned paste model met 1 of 3 components (D6).

5.2 The search

Table 5. None of the 12 candidates met every development component. Three flip the shortcut in thyroid Trap A and are not eligible (backdoor adjustment, product of experts, logit adjustment); the contrastive methods lose disease signal. The three kept met 27–28 of 29 with no loss — but round 8’s `mask_bal` also scored 28 of 29 on this development criterion and still lost on ISIC 2020, which the search could not see.

Table 5: The exploratory search: twelve candidates on development data only (development seeds, validation folds of the unaltered training sets; never a test set used for confirmation, never ovary). Round 8’s candidates are shown on the same development criterion for comparison.

Search candidate (frozen masked features)	Development components met	Losses	Eligible	Registered in round 9
backdoor adjustment	22 of 29	6	no (flips the shortcut)	–
backdoor adjustment, full image	22 of 29	1	yes	–
constraint + backdoor	23 of 29	4	yes	–
constraint + backdoor, full image	23 of 29	1	yes	–
product of experts	28 of 29	1	no (flips the shortcut)	–
group logit adjustment	28 of 29	1	no (flips the shortcut)	–
moment matching	26 of 29	1	yes	–
conditional adversarial debiasing	27 of 29	0	yes	1st
Correct-n-Contrast	20 of 29	3	yes	–
counterfactual contrastive	19 of 29	4	yes	–
V-REx	28 of 29	0	yes	3rd
IRMv1	28 of 29	0	yes	2nd
round 8 <code>mask_cmc</code> (reference)	27 of 29	1	yes	–
round 8 <code>mask_bal</code> (reference)	28 of 29	0	yes	–

5.3 Round 9: the same result on new data

Table 6 and Figure 1. Conditional adversarial debiasing, first in the sequence, met only 27 components; the sequence stopped. IRMv1 met 41, losing only ISIC 2020 overall (-0.009 [$-0.013, -0.005$], just beyond the margin of -0.01) while gaining on its conflicting pairs ($+0.028$ [$+0.022, +0.035$]); V-REx met 40. Refitted on the new data with its frozen recipe, `mask_cmc` met 41 and again lost only ISIC 2020 overall (-0.015 [$-0.024, -0.006$]): the round-8 result replicates. IRMv1 gains less than `mask_cmc` where masking fails (Table 7), so it sits at a different point of the same trade-off.

Table 6: Round 9 on new confirmation data (seeds 9101–9505): no candidate dominates masking. Round 8’s two best candidates, refitted with their frozen recipes, are descriptive references on the same data.

Candidate	Components met	All of D2 met	Losses	Role
conditional adversarial debiasing (mask_condadv)	27 of 42	no	dermoscopy, MSK held out (D1), ISIC 2019 → 2020 (D1), ovary (held out) (D3), ovary (held out) (D3), thyroid+capsule+ovary+isic (D3)	tested, not rejected
IRMv1 (mask_irm)	41 of 42	yes	ISIC 2019 → 2020 (D1)	not tested in the sequence
V-REx (mask_vrex)	40 of 42	yes	ISIC 2019 → 2020 (D1)	not tested in the sequence
class-conditional mean constraint (mask_cmc)	41 of 42	yes	ISIC 2019 → 2020 (D1)	reference (round 8)
partial reweighting (mask_bal)	38 of 42	yes	thyroid split (D1), dermoscopy, MSK held out (D1), ISIC 2019 → 2020 (D1)	reference (round 8)

Table 7: Round 9: gains over masking where masking fails (Trap A min(reversed, correlated); conflicting pairs), crossed 95% CIs.

Where masking fails	IRMv1	V-REx	mask_cmc (ref.)	mask_bal (ref.)
thyroid	+0.229 [+0.205, +0.255]	+0.322 [+0.303, +0.343]	+0.333 [+0.314, +0.353]	+0.393 [+0.363, +0.414]
capsule	+0.030 [+0.008, +0.053]	+0.156 [+0.135, +0.181]	+0.143 [+0.118, +0.172]	+0.164 [+0.140, +0.189]
ovary (held out)	+0.057 [+0.037, +0.080]	+0.081 [+0.055, +0.108]	+0.094 [+0.066, +0.122]	+0.134 [+0.101, +0.170]
dermoscopy hair	+0.120 [+0.096, +0.143]	+0.193 [+0.170, +0.216]	+0.205 [+0.181, +0.229]	+0.218 [+0.194, +0.245]
conflicting pairs, thyroid split	+0.049 [+0.030, +0.073]	+0.088 [+0.061, +0.114]	+0.062 [+0.030, +0.088]	+0.046 [+0.030, +0.063]
conflicting pairs, dermoscopy, BCN held out	+0.017 [+0.013, +0.022]	+0.022 [+0.020, +0.025]	+0.037 [+0.034, +0.039]	+0.028 [+0.021, +0.035]
conflicting pairs, dermoscopy, MSK held out	+0.036 [+0.025, +0.048]	+0.059 [+0.050, +0.067]	+0.055 [+0.026, +0.078]	+0.049 [+0.038, +0.060]
conflicting pairs, ISIC 2019 → 2020	+0.028 [+0.022, +0.035]	+0.043 [+0.037, +0.050]	+0.056 [+0.026, +0.079]	+0.039 [+0.030, +0.048]

ISIC 2020 is where masking itself is strong Table 8. Every arm loses there overall, from -0.009 to -0.029 — including the unmasked model (ERM, -0.014), whose loss comes mostly from the same-artifact pairs (-0.010). On new patients at natural prevalence, masking ranks disease better than any alternative we tested, and a method that removes the in-lesion hair shortcut additionally gives up the easy pairs.

Table 8: ISIC 2019 → 2020, round 9: the all-pairs difference to masking split exactly into the contributions of the pair types (share of pairs × difference). Every arm, including the unmasked model (ERM), loses mostly on the same-artifact pairs, i.e. disease ranking, not only on the easy pairs.

Arm	Easy pairs	Conflicting pairs	Same-artifact pairs	All pairs
erm	-0.005	0.002	-0.010	-0.014
mask_bal	-0.007	0.002	-0.010	-0.015
mask_cmc	-0.011	0.003	-0.007	-0.015
mask_condadv	-0.006	0.000	-0.024	-0.029
mask_irm	-0.005	0.002	-0.005	-0.009
mask_vrex	-0.009	0.002	-0.006	-0.012

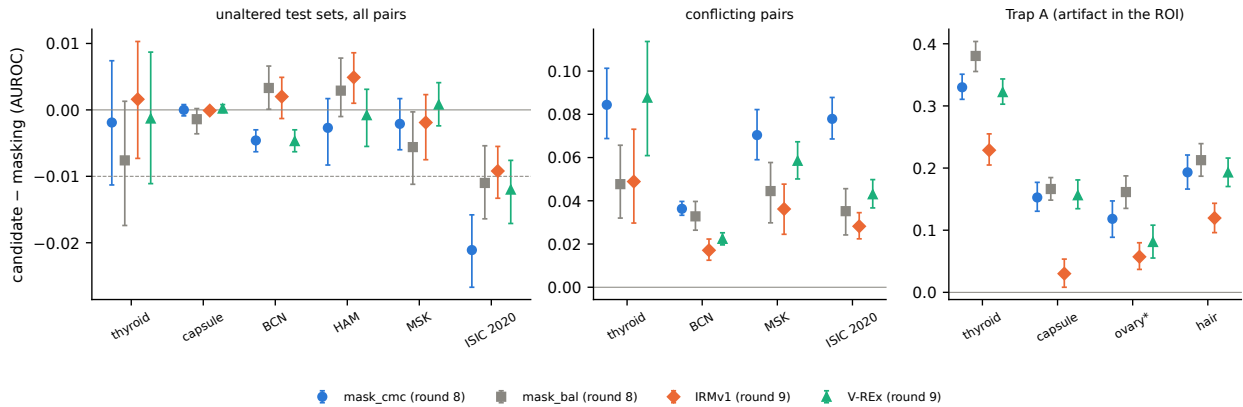


Figure 1: The four best candidates minus masking (crossed 95% CIs). Left: all pairs of the unaltered test sets (dashed: the registered margin -0.01). Middle: conflicting pairs. Right: Trap A min(reversed, correlated); *ovary held out from every choice.

6 What failed or changed

- **No dominance** in either round; the best candidates fail on one cell, ISIC 2020 overall.
- **The paste candidates** were not a fair test (too few recipients to balance the traps; pasted artifacts distinguishable from real ones).
- **The development criterion** of the search could not see ISIC 2020; the candidate it ranked first (conditional adversarial debiasing) failed most in confirmation.
- **A premise in the approval of round 8 was wrong:** the trap validation split does not carry the training association. The registered guard was kept because it is right either way.
- **Scope:** frozen DINOv2 features for the dominance test; replication in two encoders for Trap A only; the fine-tuned route (D6) was tested only with the paste candidate.

7 Anticipated questions

Is a loss of 0.01–0.02 AUROC on one external set a reason to keep masking? It is a trade-off, not a verdict. `mask_cmc` gives up -0.021 $[-0.027, -0.016]$ overall on ISIC 2020 and gains $+0.078$ $[+0.069, +0.088]$ on the cases in which the shortcut points the wrong way, and it removes masking’s failure wherever the artifact lies inside the ROI. Which matters more is a clinical choice; the registered criterion counts the overall loss as a failure, and we report it as one.

Why was the criterion so strict? Because “better than masking” is only useful if it holds everywhere a user might deploy the method. The strictness also makes the negative result informative: the candidates fail by one cell, not by many.

Were twelve search candidates a fishing expedition? The search used development data only, every attempt is logged with its prediction, and the three candidates it produced were confirmed on data generated after their registration — where they did not dominate.

Is `mask_cmc` new? No. It is an adaptation of counterfactual invariance (a class-conditional constraint) applied after masking; what is new is deriving it from the location law and the pair decomposition, and testing it this way.

Why does even the unmasked model lose on ISIC 2020? Masking is strong there: the unmasked model loses mostly on the same-artifact pairs, i.e. it ranks disease worse when the surroundings are kept, presumably because skin context and hair beside the lesion transfer poorly between the two collections. That is consistent with the location law — masking helps where the shortcut lies outside the ROI — and it explains why an alternative that keeps context, or that also removes the in-lesion shortcut, falls short of masking overall on this set.

8 Conclusion

Not supported. No remedy dominates masking (seven registered candidates in rounds 8 and 9; twelve searched). **Supported, as a trade-off.** A class-conditional mean constraint on masked features, and IRMv1, are better than masking wherever masking fails — every Trap A including a held-out cohort, the conflicting cases of every unaltered test set — and equal to it elsewhere, except for a loss of 0.01–0.02 overall AUROC on the external ISIC 2020 set, where every alternative to masking loses. In the controlled sweeps both round-8 winners dominate masking.

9 Questions for the supervisor

1. Should the paper recommend `mask_cmc` as the remedy to use when artifacts may lie inside the ROI, stating its ISIC 2020 cost, or only report it?
2. Should the negative dominance result (two registered rounds) enter the main text, as the answer to “so what should we use instead?”?
3. Is closing the remedy question here — no further search — acceptable?

References